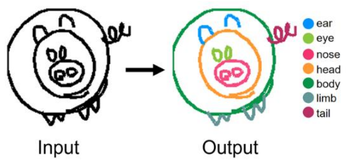
Context-Enhanced Sketch Segmentation with BiLSTM and GCN

Team 17

Introduction



Sketch context segmentation groups freehand strokes into semantic parts.

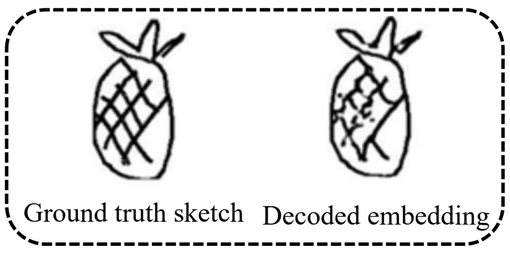
Prior work (ContextSeg) introduced the use of a **distance map**, achieving state-of-the-art results. However, because it relies only on **geometry-based features**, it struggles on cluttered sketches where overlapping and interleaved strokes confuse the embedding. We address this with the proposed ***SeqSpa module***, a lightweight relation block that captures **stroke–stroke context** through **drawing order** and **overlapping information**, yielding more robust segmentation under clutter.

Problem

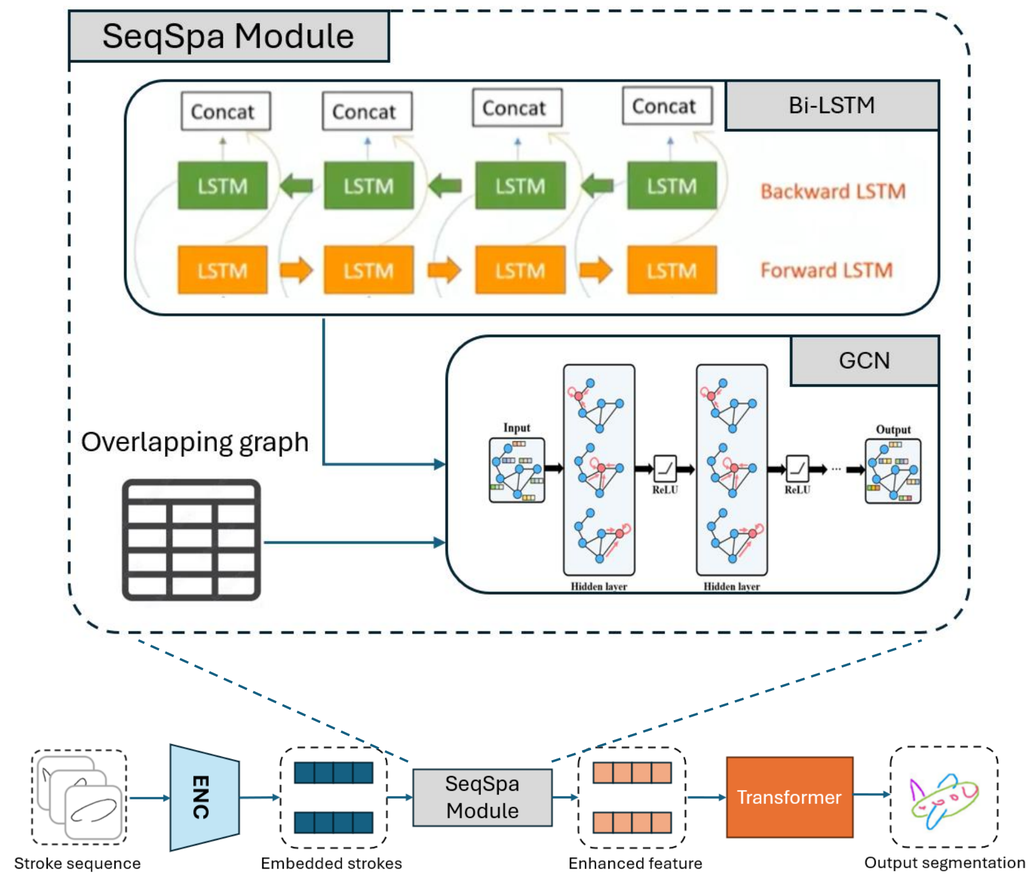
Prior work (ContextSeg) encodes each stroke using **geometric features**—an embedding learned from its **raster image** together with a **distance map** *(to better preserve thin strokes)*, plus **absolute (x,y)** coordinates—then uses a Transformer with a pointer network to assign strokes into groups. While effective on clean sketches, this geometry-only design struggles under clutter, where **overlapping and interleaved** strokes cause embeddings to lose semantic meaning.

Stroke Distance Field

$$DF^s(s_i) = \frac{1}{1 + k * e^{dis^p_{s_i}}}$$



Methodology



Sequential Encoder – BiLSTM

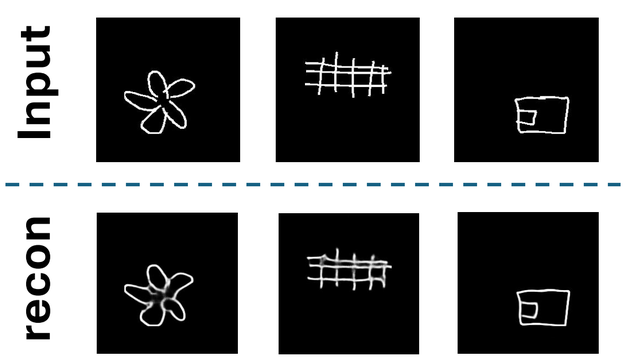
- **What it does:** Uses a bidirectional LSTM to capture the temporal relation between strokes.
- **Why it helps:** Cluttered strokes are often drawn **in sequence**, so drawing order provides a strong cue for disambiguation.

Spatial Encoder – GCN

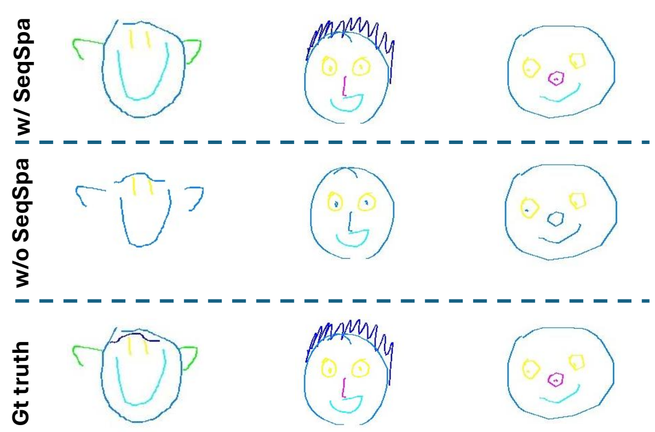
- **What it does:** Builds a graph of overlapping strokes and applies a Graph Convolutional Network to propagate spatial context.
- **Why it helps:** Cluttered sketches often contain **overlapping strokes**, and modeling these relations improves grouping accuracy.

Experiment

reconstruction result



Visual comparison



Statical comparison

Category	Reproduced			w/ SqeSpa			ContextSeg		
	SAcc	GAcc	CAcc	SAcc	GAcc	CAcc	SAcc	GAcc	CAcc
Basket	96.7	95.4	92.2	96.5	**97.3**	**95.2**	97.8	98.0	94.2
Calculator	98.0	96.9	95.4	96.8	**97.8**	**97.0**	99.2	98.7	97.5
Candle	97.5	96.2	94.5	**97.9**	**98.5**	**97.2**	98.7	98.5	96.0
Face	93.0	81.8	79.5	92.4	**97.9**	**91.3**	96.4	98.4	94.8
Pineapple	96.8	96.3	95.5	**97.4**	**97.8**	**97.2**	97.3	97.0	92.5

Results

- **Consistent gains across categories**
- **Stronger benefits in complex cases**
- **Robustness under clutter**

Contribution

Downloadable Code

We rebuilt the code in PyTorch, making it more current and easier to extend for new ideas.

Why is this topic useful?

Practical work with sketch segmentation

- Free2CAD: Parsing Freehand Drawings into CAD Commands *SIGGRAPH 2022*
- Fine-Grained Sketch-Based Image Retrieval-*CVPR 2020*
- DeepSketch2Face: A Deep Learning Based Sketching System for 3D *SIGGRAPH 2017*

Other

- **Quick, Draw!** -By Google Creative Lab
- LLM models, like **Chatgpt**, **Claude**, **Gemini** can recognize sketch.